CS304P - Quiz No.04

Date: 08 January 2026

Q.1 In case of public inheritance between Base class A and Derived class B, which of the folllowing data members of Base class will be hidden in Derived class?

A. public

B. protected

C. private

D. both public and private

Q.2 What is the purpose of explicit specialization in templates?

A. To handle specific data types that the template cannot handle successfully

B. To simplify the template syntax

C. To avoid the use of templates

D. To improve runtime performance

Q.3 For the given class definition, which one is correct? #include<iostream> using namespace std; class A{ int x,y; public: int Add(int a,int b){return a+b;} int Add(int c){return c;} };

A. None of the given options

B. Function Add is overloaded

C. Function Add is overridden

D. Function Add has an error

Q.4 The partial specialization pattern T* says that this specialization is to be used for every _____.

A. Meta type

B. Data type

C. Virtual type

D. Pointer type

Q.5 class A { public: int Test(int x) {return x*x;} int Test(int x,int y) {return x*y;} }; class B : public A { public: int Test(int x) { return x*x*x;} }; In the above class, the method Test is_____.

A. Both

B. Overloaded

C. Overridden

D. None of the given options

Q.6 Which of the following is the correct syntax to declare a vector of type int?

A. vector {int} sampleVector;

B. sampleVector vector<int> ;

C. vector<int> sampleVector;

D. int <vector> sampleVector;

Q.7 How can you define a class template in C++?

A. template<name T> class MyClass { ... };

B. template<class T> MyClass { ... };

C. template<type T> class MyClass { ... };

D. template<typename T> class MyClass { ... };

Q.8 Suppose you have a function: void sum(); Choose correct syntax from the given options which will make this function pure virtual function.

A. void sum()=0;

B. virtual void sum()=0;

C. void sum() virtual=0;

D. sum()=0;

Q.9 Which of the given is not a component of STL?

A. Container

B. Iterator

C. Cursor

D. Algorithm

Q.10 Which of the following is the correct syntax for creating a function Test(T) friend of a template class?

A. void Test(T);

B. friend void Test(T);

C. void friend Test(T);

D. void Test(T) friend;

Q.11 In a function template template<typename T>, what does T represent?

A. Non-type parameter

B. Built-in type

C. Function return value

D. Type parameter

Q.12 What feature enables same-named template functions to exist?

A. Inheritance

B. Function overloading

C. Polymorphism

D. Specialization

Q.13 Which one of the following is an example of complete specialization?

A. template <typename T> class Vector<T*> { }

B. template <typename T> class Vector { }

C. template <class T> class Vector<T *> { }

D. template<> class Vector <char*> { }

Q.14 What does the compiler check first during template resolution?

A. Partial specialization

B. General template

C. Function overloading

D. Complete specialization

Q.15 Which of the following is the correct syntax for a function template in C++?

A. template void func<class T>(T a);

B. template<class T> void func(T a);

C. template<type T> func(T a);

D. func<class T>(T a);

Q.16 If a function template has parameters `T` and `U`, calling `func(1, 2.5)` results in:

A. Both deduced as double

B. Both deduced as int

C. Compilation error

D. T=int, U=double

Q.17 Which of the following is an example of partial specialization?

A. template<class T, float>

B. template<int>

C. template<class T, class U>

D. template<int , float , char>

Q.18 If Vector<char* > is instantiated, which version will the compiler choose?

A. Function overloading

B. General template

C. Partial specialization

D. Complete specialization

Q.19 Which of the following is not the correct type of inheritance in C++?

A. public

B. protected

C. secret

D. private

Q.20 virtual void print(); is an example of_____.

A. simple function

B. pure virtual function

C. template function

D. virtual function
